

NON-TECHNICAL MODEL SUMMARY REPORT

1. Purpose of the model

The HealthConnect Data Science model was developed to identify appointments with a higher likelihood of becoming a no-show. It uses information about appointments and previous appointment history to produce a structured risk signal that could support appointment-management decisions.

2. Final model

The final candidate is a Logistic Regression model using 13 predictor features. The model was selected after comparison with Random Forest, cross-validation, error analysis, segment testing, refinement and final reproducibility checks.

3. Key performance results

Measure	Final result
Accuracy	61.50%
Precision	60.87%
Recall	69.28%
F1 Score	64.80%
ROC-AUC	66.47%

The model correctly identified 336 of 485 actual no-shows in the test data, which is approximately seven out of every ten actual no-shows. However, it also produced 216 false positives and missed 149 actual no-shows.

4. Key business findings

Booking lead time and previous no-show history were particularly relevant for understanding no-show risk. These patterns were also supported by findings from the Data Analytics track. The model therefore demonstrates that existing appointment information contains useful predictive signals, although the signal is not strong enough to support fully automated decisions.

5. Business value

The model could potentially support appointment planning by helping identify appointments that may require additional attention. Its appropriate role is as a decision-support tool that complements operational judgement rather than replacing human decision-making.

6. Limitations and responsible use

- Model performance remains moderate, and both false positives and false negatives are present,
- The HealthConnect dataset is fictional and was developed for an educational project; the model has not been validated in a real healthcare environment,
- The identified relationships should not be interpreted as proof that particular factors cause patients to miss appointments,
- The model should not be used as the sole basis for decisions about individual patients.

7. Recommended next steps

- Validate the model using real operational appointment data,
- Evaluate and optimise the classification threshold according to the practical costs of false positives and false negatives,
- Assess and calibrate predicted probabilities before operational use,
- Perform external and temporal validation to determine whether performance remains stable across different periods and populations,
- Complete downstream ML Engineering integration and end-to-end workflow validation when collaboration becomes available.

8. Conclusion

The HealthConnect model provides a reproducible proof of concept for predicting appointment no-show risk. It demonstrates how appointment and patient-history information can be converted into a useful risk signal, while also showing the importance of validation, error analysis and responsible interpretation. At this stage, the model should be considered a tested decision-support prototype rather than a production-ready healthcare system.